

Modeling and Control of a half Car Active Suspension System using Sliding Mode Controller and Linear Quadratic Regulator Controller

Abdussalam Ali Ahmed
Mechanical Engineering Department
Bani Waleed University
Bani Waleed, Libya
abdussalam.a.ahmed@gmail.com

Mohamed El Malki
Department of Physics, Faculty of Sciences
Mohammed First University
Oujda 60000, Morocco
m.elmalki@ump.ac.ma

Hisham.F. A. Ramadhan
Control System Engineering
College of Electronic Technology
Bani Walid, Libya
sq1r8t@gmail.com

Yasser Fathi Nassar
Mechanical and Renewable Energy Eng. Dept.,
Wadi Alshatti University
Brack, Libya
y.nassar@wau.edu.ly

Abstract— This paper illustrates the use of two distinct control strategies. Namely, the Sliding Mode Control (SMC) and a Linear Quadratic Regulator (LQR) controller to monitor and handle the suspension system vibrations that occurred. We used a half car suspension model with (4DOF). The necessary suspension performance attributes, such as ride comfort, suspension deflection and car handling require a comprehensive control system. Three components make up this control system, which is carried out using the MATLAB application: the controller, the suspension system model, and input signals (road profile and actuator force). A comparison of the half car active suspension system LQR and SMC controllers as well as the uncontrolled suspension system is presented.

Keywords: *Sliding Mode Control, Active Suspension, Half Car Model, Linear Quadratic Regulator.*

I. INTRODUCTION

Recently, vehicles with active suspension have been developed in the effort of increasing the performance and efficiency of vehicles suspension, and the process is moving toward an additional improvement. [1].

A suspension system on a vehicle usually performs the following important functions: (i) isolating vehicle's body from road vibration to ensure a smooth ride, (ii) to ensure an excellent road holding, (iii) to ensure proper handling [2].

The active suspension systems, semi-active/passive and suspension systems are the three basic categories of automotive suspension systems. The vehicle body and its wheels have springs and other suspension system components positioned between them. The suspension components enable the forward adjustment of the suspension stroke deviation. The suspension stroke is constrained by the characteristics of the chassis to a certain range of values.

When the suspension deviation hits these defined limited levels, the ride comfort is degraded [3].

In active suspension, a force actuator is positioned between the body of the vehicle and the wheels. The actuator is managed by the feedback controller. A regulated suspension system enables forward correction between the performance standards for the suspension aberrations and ride comfort [4-5].

The car's active suspension has recently been the subject of numerous studies. Improvements by researchers have been done to enhance ride comfort and road handling. In this regard, various control techniques have been used such as PID control of a nonlinear half-car active suspension system by force feedback [6], optimal fuzzy control design for half-car active suspension systems [7], half-car active suspension system controlled using model predictive control [8]. In addition, Half-Car active suspension can be controlled by H_∞ Method [9], and by Adaptive Harmonic Control Method [10]. The Sliding Mode Control [11] were incorporated to the active vehicle suspension field of half-car active suspension. There are a number of control technologies that can enhance the performance attributes of an active suspension system, these techniques, however, come with certain implementation challenges and a specialized performance decision schedule or complicated learning process.

In this study, an examination of the vehicle active suspension performance by the LQR and SMC controllers.

This study is organized as following: section 2 describes the road profile modeling as well as the theoretical model of the half car suspension. Section 3 briefly presents the LQR method and SMC controller. Section 4 compares the two controllers used in the Simulink of the active suspension system of the half car. A discussion of the results is presented in section 5 followed by a conclusion in section 6.

II. MATHEMATICAL MODELING

The (4DOF) half car suspension model is presented in Fig. 1, while Table 1 presents the parameters considered in this study. This particular suspension model is the most used model, this is crucial when researching vehicle suspension, primarily road handling and ride comfort attributes. The parameters describe the motion vibration of the vehicle's body/wheels. The suspension system's primary components of are dampers C_{sf} , C_{sr} and springs k_{sf} , k_{sr} , in addition to forces actuators f_{sf} , f_{sr} . In case of passive suspension, the actuator force value must be vanished. m_s symbol indicates the sprung mass that refer to a half of vehicle's mass. The unsprung mass m_{uf} , m_{ur} refer to the mass of the wheels, respectively. The damper coefficients are indicated by the symbols C_{sf} , C_{sr} , while k_{uf} , k_{ur} being the vertical stiffness of the tires. The road profile is indicated by the symbols W_{sf} , W_{sr} and unsprung mass' symbols indicated by m_{uf} , m_{ur} and sprung mass indicated by m_s .

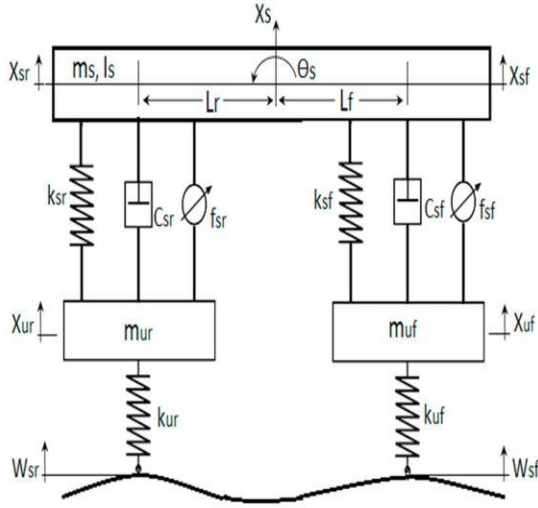


Fig. 1. Model of the suspension of a half car.

TABLE 1. Parameter's values of the half car suspension.

Parameter	Value/Unit
Sprung mass, m_s	1795 (kg)
Moment of Inertia, I_s	3443 (kg m ²)
Space between front and rear suspension, L	2.99 (m)
Space between center of gravity and front suspension, L_f	1.27 (m)
Space between center of gravity and rear suspension, L_r	1.72 (m)
Unsprung mass, m_{uf}	87.2 (kg)
Unsprung mass, m_{ur}	140.1 (kg)
Front Suspension stiffness, k_{sf}	66820 (N/m)
Rear Suspension stiffness, k_{sr}	18620 (N/m)

Front tire stiffness, k_{uf}	101120 (N/m)
Rear tire stiffness, k_{ur}	101120 (N/m)
Front Suspension damper, C_{sf}	1190 (Ns/m)
Rear Suspension damper, C_{sr}	1000 (Ns/m)

The motion equations of the system is given by second law of Newton as follows:

$$m_s \ddot{x}_s + C_{sf}(\dot{x}_{sf} - \dot{x}_{uf}) + k_{sf}(x_{sf} - x_{uf}) + C_{sr}(\dot{x}_{sr} - \dot{x}_{ur}) + k_{sr}(x_{sr} - x_{ur}) - f_{sf} - f_{sr} = 0 \quad (1)$$

$$I_s \ddot{\theta}_s + L_f(C_{sf}[\dot{x}_{sf} - \dot{x}_{uf}] + k_{sf}[x_{sf} - x_{uf}] - f_{sf}) - L_r(C_{sr}[\dot{x}_{sr} - \dot{x}_{ur}] + k_{sr}[x_{sr} - x_{ur}] - f_{sr}) = 0 \quad (2)$$

$$m_{uf} \ddot{x}_{uf} - C_{sf}(\dot{x}_{sf} - \dot{x}_{uf}) - k_{sf}(x_{sf} - x_{uf}) + k_{uf}(x_{uf} - w_{sf}) + f_{sf} = 0 \quad (3)$$

$$m_{ur} \ddot{x}_{ur} - C_{sr}(\dot{x}_{sr} - \dot{x}_{ur}) - k_{sr}(x_{sr} - x_{ur}) + k_{ur}(x_{ur} - w_{sr}) + f_{sr} = 0 \quad (4)$$

where (1) is the equation of vertical movement at the vehicle body (2) is the pitch angle movement at the vehicle body, and (3)-(4) are respectively the vertical movement at unsprung mass.

$x_{sf} - x_{uf}$ represents the front suspension deflection, $\dot{x}_s$ being the sprung mass speed, $\ddot{x}_s$ indicates the body of the car acceleration (comfortable comfort indicator), $x_{uf} - W_{sf}$ is the deflection of the front tire (indicator of the road-handling), while $\dot{x}_{uf}$ is the velocity of the front tire. The rear suspension can be represented respectively for suspension deflection and road-handling by replacing sub f with sub r .

A representation of the System can be given by the following state space matrix form:

$$\dot{x} = Ax + Bu + Dist * w = Ax + [B \quad Dist] \begin{bmatrix} u \\ w \end{bmatrix}$$

$$y = Cx$$

In what follows, a representation of the system with state space Simulink block will be used to minimize complexity and wiring (Fig. 4, and Fig. 5).

• Road Profile

The input road bumps for the system is represented by the Simulink model as shown in Fig. 2.

A single bump road input (*front_dist*) shifted 0.1m and labeled as (*rear_dist*) as shown in Fig. 3 is applied to transmit road status and conditions, as well as to validate the designed control system. This road disturbance, also known as a profile of the road, is intended to reach a certain bump height of 0.1m as shown in Fig. 3.

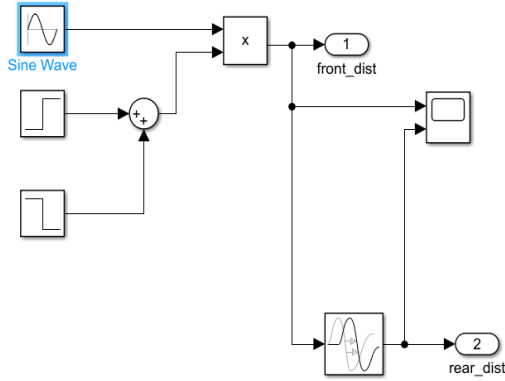


Fig. 2. Model of the Bumpy road input subsystem.

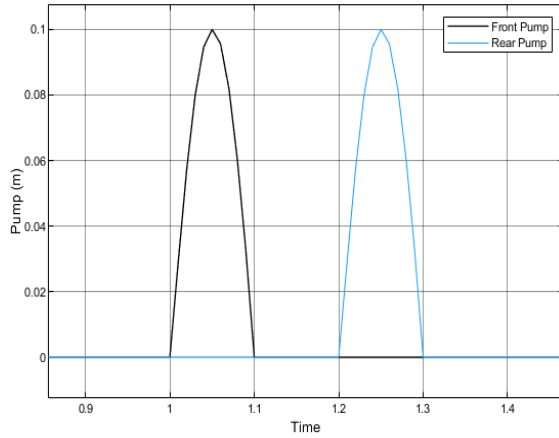


Fig. 3. Bumpy road input subsystem Simulink.

III. STRATEGY OF THE CONTROL

This section is composed of two parts. The first part briefly describes the controller performed in this work. The second part shows the Simulink model structures representation for the active system suspension for the controller.

• Controller Design Using LQR

The control technique LQR is largely used in various researches background. This technique was applied to many car models. Namely, half-car models, basic quarter-car models, and even full-car models. The LQR technique has the advantage of allowing the engineers' manipulating the performance factor index to manipulate control weights, so control cost can be varied.

When building a LQR controller, optimal control is achieved by selecting the control signal $u(t)$ in order to reduce the quadratic cost function. The theoretical formula of this latter is expressed as:

$$J = \int_0^{\infty} (x(t)^T Q x(t) + u(t)^T R u(t)) dt \quad (1)$$

The state vector is indicated by the parameter x , and u is the control vector. $(n \times n)$ and $(r \times r)$ must be positive and symmetric definite weighted Matrices. The key matrix condition Q and R need to be positive definite ($R = R^T \geq 0$ and $Q = Q^T \geq 0$).

The control cost values given by the weighted matrix R and state cost Q are determined by the designer depending on its preferences.

The full state feedback system configuration is illustrated in Fig. 4.

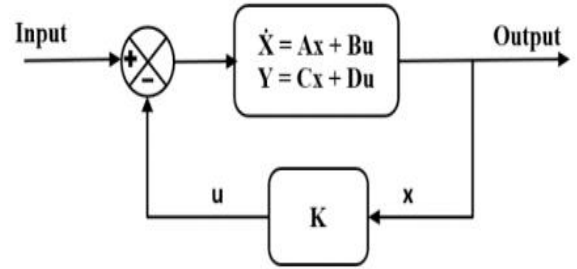


Fig. 4. Full State feedback.

The most used state feedback rule control signal,

$$u(t) = -K(t)x(t) \quad (2)$$

K being the matrix of the feedback gain for the LQR controller. Its measured from:

$$K = R^{-1} B^T P \quad (3)$$

The Riccati algebraic equation is used to calculate the P matrix as follows:

$$A^T P + AP + PBR^{-1}B^T P + Q = 0 \quad (4)$$

By selecting the Q matrix given by:

$$Q = \begin{bmatrix} 1000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 100 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1000 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 100 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 100 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1000 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 100 \end{bmatrix}$$

where $R = 0.00001$.

Then, from (7)-(8) value of K can be calculated and given:

$$K = \begin{bmatrix} 10181 & -1782 & 76 & -31 & 743 & -19588 & 141 & 1294 \\ -61 & -31 & 9708.4 & -1820 & -450.3 & 1137.9 & 2515.5 & -31909 \end{bmatrix}$$

The LQR controller is embedded in the Simulink model for the half car suspension model presented in Fig. 5.

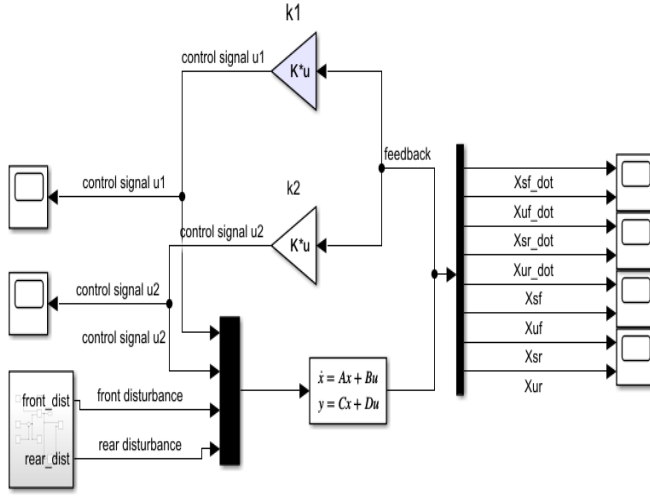


Fig. 5. Active suspension system with LQR controller.

• Sliding Mode Control

The sliding mode control (SMC) is a nonlinear control technique featuring important properties of robustness, easy tuning, and accuracy. SMC controller is based on a two-phase controller design. The first stage engages designing a sliding surface where the sliding motion matches to design requirements while the second stage is involved with choosing a law of control that attracts the states of the system to the switching surface.

$$u = \ddot{u} + u_{uncer} \quad (5)$$

$\ddot{u}$ is constructed from the first phase which is sliding surface design the sliding surface design mostly depends on the error e along with a certain number of its derivatives.

$$\sigma = \sigma(e, \dot{e}, \dots, e^{(k)}) \quad (6)$$

The sliding surface conditions to be designed that σ should be select as $\sigma = 0$ when the system error $e = 0$. Also the sliding surface first derivative $\dot{\sigma}$ must contain the control signal $\ddot{u}$ a standard form can be used

$$\sigma = \left(\frac{d}{dt} + \lambda\right)^n (x - x_d) = \left(\frac{d}{dt} + \lambda\right)^n e \quad (7)$$

Where $n = r - 1$, r being the order of system, x_d is state desired value, and λ is a number tuned by the designer to change system performance. The equation (11) will ensure that if $\sigma = 0$, then $e = 0$.

From (9) the u_{uncer} has various ways to be designed the easiest way is to consider u_{uncer} as follows:

$$u_{uncer} = -U * sgn(\sigma) \quad (8)$$

Where U is a number tuned by the designer to change system performance, $sgn(\sigma)$ can be represented as

$$sgn(\sigma) = \begin{cases} -1 & \sigma < 0 \\ 1 & \sigma > 0 \end{cases} \quad (9)$$

However, the sgn function can be replaced with sat function to minimize chattering problem.

The sliding surface design for the half suspension for the front portion of the system is similar to the rear portion, so the front portion is only shown, starting by using the (11) formula:

$$\sigma_f = \dot{x}_{sf} - \dot{x}_{sfd} + \lambda_{sf}(x_{sf} - x_{sfd}) + \dot{x}_{uf} - \dot{x}_{ufd} + \lambda_{uf}(x_{uf} - x_{ufd}) \quad (10)$$

By differentiating σ_f should get us $\ddot{u}$ the first part of the control u :

$$\dot{\sigma}_f = \ddot{x}_{sf} + \lambda_{sf}\dot{x}_{sf} + \ddot{x}_{uf} + \lambda_{uf}\dot{x}_{uf} \quad (11)$$

$\ddot{x}_{sf}$, $\ddot{x}_{uf}$ contain $\ddot{u}$ which is f_{sf} the front actuation force.

The second stage of the design is getting u_{uncer} from is going to be:

$$u_{uncer} = -U * sat(\sigma) \quad (12)$$

The SMC controller is embedded in the Simulink model for the suspension system of the car as presented in Fig. 6.

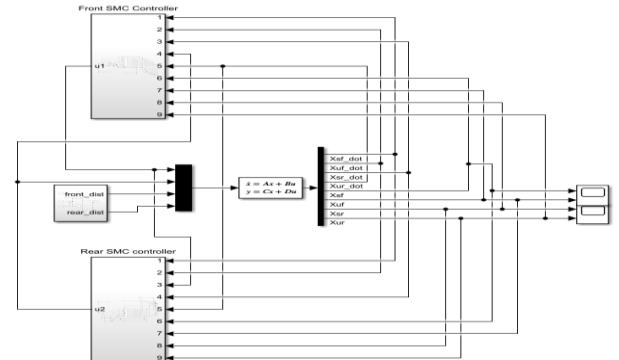


Fig. 6. Half-car active suspension model with SMC.

The front SMC controller Simulink model is illustrated in Fig. 7.

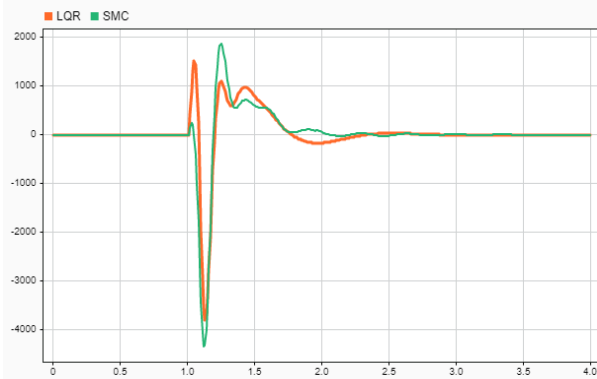


Fig. 13. The front control signal force.

The following section of the paper has been devoted to discussing and explaining what has been performed in order to acknowledge the outcome results and output of the system to ensuring the success of the used controllers.

V. RESULTS AND DISCUSSION

The LQR control technique and SMC are compared to the uncontrolled system to investigate the active suspension system. Fig. 8 compares the suspension deflection response, while Fig. 9 shows the deflection of the tire which is an indicator of the road handling. Clearly, a less overshoot and a longer settling time obtained in the LQR controller compared to the SMC controller.

Fig. 10-11 demonstrate the unsprung and sprung mass velocities for the half car suspension front. In the SMC case, there is less settling time and a lower value for the sprung mass velocity compared to the LQR case.

Additionally, the two controllers enhanced the sprung mass's acceleration as shown in Fig. 12. Obviously, SMC control method gives better performance in the acceleration curves. Given that it serves as a measure of ride comfort, the acceleration parameter of the sprung mass is given significant weight when designing car suspension systems.

The control signals from Fig. 13 shows how SMC consumes more power compared with LQR, power consumption important for real world implementation consideration, more power means less applicable controller.

VI. CONCLUSION

This work describes the half car active suspension using a mathematical equation based on two control methods. The objective of this study is to analyze the performance of the system (SMC and LQR). The half-car suspension model has been implemented in Simulink models by MATLAB. Our results show that both control methods measures and regulate the suspension system and produce a good performance system, although the SMC technique produces slightly better system performance than the LQR controller. To summarize this portion of the study, many control techniques and intelligent controllers had previously been

utilized in multiple research, and gave an excellent results. However, to separate this work from previous research, it can be stated that this study employed two intelligent mode of control and compared their results in terms of providing an adequate efficient suspension system. This can be considered as a positive contribution that is done via this research.

REFERENCES

- [1] Sang Bong Kim, Tan Tien Nguyen and Van Giap Nguyen, (2000). "Control of Active Suspension System," *Transaction on Control, Automation, and Systems Engineering*, (2)1.
- [2] R. Rajamani, *Vehicle dynamics and control*, Business Media & Springer Science, 2011.
- [3] Sheilza Jain, Swati Gaur, (2013) "Vibration Control of Bus Suspension System using PI and PID Controller," *International Journal of Advances in Engineering Sciences*, July 2013.
- [4] F.J. D'Amato and D.E. Viassolo, (2000). "Fuzzy control for active suspension," *Mechatronics*, 10, 897–920.
- [5] Yao, G. Z., F. F. Yap, G. Chen, W. H. Li, and S. H. Yeo, 2002. "MR damper and its application for semi-active control of car suspension system." *Mechatronics*, 963-973.
- [6] John E. D. Ekoru; Olurotimi A. Dahunsi; Jimoh O. Pedro, (2011). "PID control of a nonlinear half-car active suspension system via force feedback." DOI: 10.1109/AFRCON.2011.6071979.
- [7] Shing-Jen Wu; Hsin-Han Chiang; Jiun-Hau Chen; Tsu-Tian Lee, (2004). "Optimal fuzzy control design for half-car active suspension systems." DOI: 10.1109/ICNSC.2004.1297504.
- [8] Jimoh O. Pedro, Sakhile M.S. Nhlapo, Lindokuhle J. Mpanza, (2020). "Model Predictive Control of Half-Car Active Suspension Systems." ISSN 2405-8963.
- [9] Karim Afshar K, Javadi A., (2019). "Constrained H^∞ control for a half-car model of an active suspension system with actuator time delay by predictor feedback." *Journal of Vibration and Control*. 25(10):1673-1692. doi:10.1177/1077546319828457.
- [10] Unguritu, Maria-Geanina, Nichitelea, Teodor-Constantin, Selişteanu, Dan. (2022). "Design and Performance Assessment of Adaptive Harmonic Control for a Half-Car Active Suspension System." *Complexity*. 2022. 10.1155/2022/3190520.
- [11] Haifa Souilem, Sonia Mahjoub, Nabil Derbel, (2016). "Sliding Mode Control of Half-car Active Suspension." *International Journal of Computer Applications*.
- [12] M. Ahmed and F. Svaricek, (2013). "Preview control of semi-active suspension based on a half-car model using Fast Fourier Transform," *10th International Multi-Conferences on Systems, Signals & Devices 2013 (SSD13)*, 1-6, doi: 10.1109/SSD.2013.6564120.
- [13] S. A. Abu Bakar, P. M. Samin, H. Jamaluddin, R. A. Rahman and S. Sulaiman, (2015). "Semi active suspension system performance under random road profile excitations," *2015 International Conference on Computer, Communications, and Control Technology (I4CT)*, 93-97, doi: 10.1109/I4CT.2015.7219544.
- [14] Ho, C.M.; Nguyen, C.H.; Ahn, K.K. (2022). "Adaptive Fuzzy Observer Control for Half-Car Active Suspension Systems with Prescribed Performance and Actuator Fault." *Electronics*, 11, 1733. <https://doi.org/10.3390/electronics11111733>.
- [15] A. A. Ahmed, A. Alsharif, T. Triwiyanto, M. Khaleel, C. W. Tan and R. Ayop, (2022). "Using of Neural Network-Based Controller to Obtain the Effect of Hub Motors Weight on Electric Vehicle Ride Comfort," *2022 IEEE 2nd International Maghreb Meeting of the Conference on Sciences and Techniques of Automatic Control and Computer Engineering (MI-STA)*, Sabratha, Libya, 189-192, doi: 10.1109/MI-STA54861.2022.9837608.
- [16] A. A. Ahmed, H. A. Eissa, A. M. Faraj, A. Albagul, M. Belrzaeg and A. Alsharif, (2021). "Suspension System Modelling And Control For An Electric Vehicle Driven by In-Wheel Motors," *2021 IEEE*

International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON), Dhaka, Bangladesh, 1-5, doi: 10.1109/RAAICON54709.2021.9929611.

- [17] C. U. Dogruer, (2022). "Optimal Mechanical Design of Half-Car Vehicle Suspension System Components," 2022 13th Asian Control Conference (ASCC), Jeju, Korea, Republic of, 2022, 1302-1308, doi: 10.23919/ASCC56756.2022.9828244.
- [18] Y. Huang, H. Hou, J. Na, H. He, J. Zhao and Z. Shi, (2023). "Output Feedback Control of Hydraulic Active Suspensions With

Experimental Validation," in IEEE Transactions on Circuits and Systems II: Express Briefs, (70)2, 631-635, doi: 10.1109/TCSII.2022.3208887.

- [19] Y. Liao, D. Ning, P. Liu, H. Du and W. Li, (2023). "A Versatile Semiactive Electrically Interconnected Suspension With Heave-Roll Vibration Decoupling Control," in IEEE Transactions on Industrial Electronics, (70)6, 6159-6169, doi: 10.1109/TIE.2022.3196359.